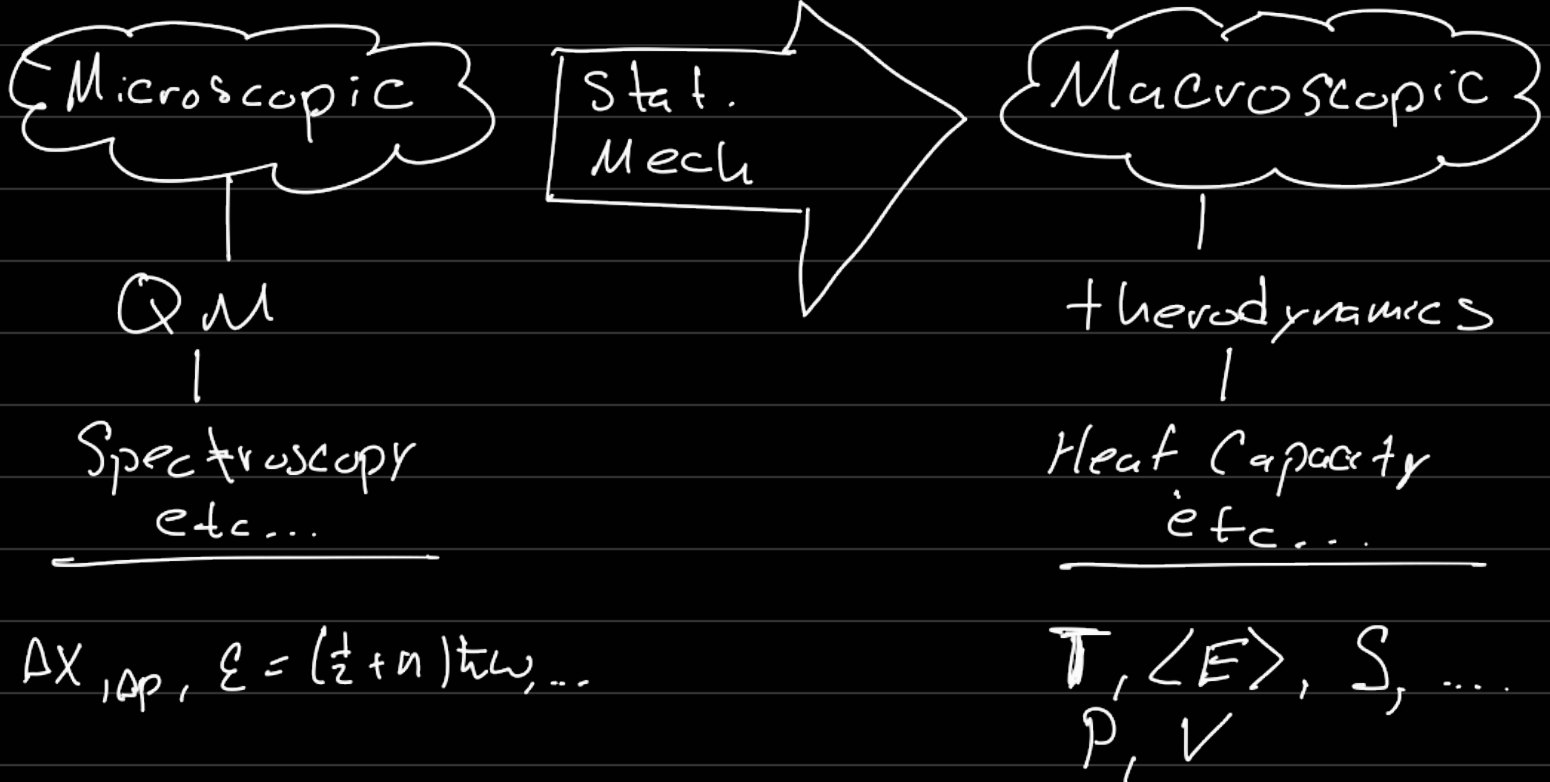




Ensemble: collection of identical systems

Micro-canonical Ensemble:

- Systems with fixed E & N

Canonical Ens:

- E : Constant
- Heat bath $\partial E / \partial T \neq 0$

Grand Canonical Ens:

- System free to exchange both E & N

* Micro Canonical: Isolated System

E : fixed

$$\hat{H}|\psi\rangle = E|\psi\rangle$$

$$N \sim 10^{23}$$

Microstates: Energy eigenstates

→ Many microstates can have the same E

- System is in a mixed state

$$\langle \hat{O} \rangle = \sum_n P(n) \langle n | \hat{O} | n \rangle$$

$$\langle \hat{O} \rangle \equiv \langle 4 | \hat{O} | 4 \rangle$$

$$\begin{aligned} \langle 4 | &= \sum_{n,m} \frac{\langle 4 | 4 \rangle}{\sum_m |c_m|^2} c_n c_m \langle n | \hat{O} | m \rangle \\ &= \sum_n \left(\frac{|c_n|^2}{\sum_m |c_m|^2} \right) \langle n | \hat{O} | n \rangle \end{aligned}$$

* For an isolated system in Equilibrium, all accessible microstates are equally likely

$\Omega(E)$ = # of states with energy E

• Prob. that the system is in $|n\rangle$

$$P(n) = \frac{1}{\Omega(E)}$$

$$P(E') = 0$$

• Microcanonical: Fixed E

ex:

$$\Omega(E): N \sim 10^{23}$$

↑ or ↓

$$\Omega(E) = 2^{10^{23}}$$

$\Omega(E) \sim$ states in $E + \delta E$

Entropy: $S(E) = k_B \ln(\Omega(E))$ | $k_B \sim 5 \text{ J K}^{-1}$

$$\Omega \sim e^N \text{ so that } S \sim N$$



$$W(E_1, E_2) = W_1(E_1) W_2(E_2)$$

$$S(E_1, E_2) = S_1 + S_2$$

Second Law of thermodynamics:

Put the two systems together

- Assume they are isolated

$$E = E_1 + E_2$$

$$E_2 = E - E_1$$

$$W(E) = \sum_{\{E_i\}} W_1(E_i) W_2(E - E_i)$$

$$W_i = e^{\frac{S_i}{k}}$$

$$W(E_{\text{tot}}) = \sum_{\{E_i\}} \exp \left(\frac{S_1}{k_B} + \frac{S_2(E_{\text{tot}} - E_i)}{k_B} \right)$$

$$\bullet \quad p = 1/W(E_{\text{tot}})$$

$$S_{\text{tot}}(E_{\text{tot}}) = k_B \ln(W_{\text{tot}}) \geq S_1(E_1) + S_2(E_2)$$

$$dF = dE_A = -dE_B \implies \frac{dE_A}{dE} = 1 = - \frac{dE_B}{dE}$$

Assumes energy is lost from B $\rightarrow$ A

$$E = E^*$$

$$\bullet \quad \left. \frac{d(W_A W_B)}{dE} \right|_{E=E^*} = 0$$

$$= \frac{\partial W_A}{\partial E} W_B + \frac{\partial W_B}{\partial E} W_A = 0$$

$$\frac{dW_A}{dE} = \frac{dE_A}{dE} \frac{\partial W_A}{\partial E_A} = \frac{\partial W_A}{\partial E_A}$$

$$\frac{dW_B}{dE} = \frac{\partial E_B}{\partial E} \frac{\partial W_B}{\partial E_B} = - \frac{\partial W_B}{\partial E_B}$$

$$\frac{\partial W_A}{\partial E_A} \frac{1}{W_A} = \frac{\partial W_B}{\partial E_B} \frac{1}{W_B}$$

$$\left(\frac{\partial \ln(W_A)}{\partial E_A} \right)_V = \left(\frac{\partial \ln(W_B)}{\partial E_B} \right)_V$$

(Equilibrium)

* Statistical systems tend towards higher prob. states

$$\frac{1}{T} = k_B \left(\frac{\partial \ln(W)}{\partial E} \right)_V$$

$$S = k_B \ln(W(E, s, \dots))$$

$$\bullet S(\text{total}) \approx S_1(E_1) + S_2(E_{\text{tot}} - E_1) \\ \geq S_1 + S_2$$

$$\bullet \delta E_1 = -\delta E_2$$

$$\delta S = \frac{\partial S_1(E_1)}{\partial E} \delta E_1 + \frac{\partial S_2(E_2)}{\partial E} \delta E_2$$

$$= \left(\frac{\partial S_1}{\partial E} - \frac{\partial S_2}{\partial E} \right) \delta E_1$$

$$= \left(\frac{1}{T_1} - \frac{1}{T_2} \right) \delta E_1$$

$$\bullet \delta S \geq 0$$

$$\text{and } \delta E_1 > 0$$

$$\text{so } \frac{1}{T_1} - \frac{1}{T_2} \geq 0$$

$$T_1 \quad T_2 \quad 0$$

$$\boxed{T_2 > T_1}$$

→ Energy flows from $T_{\text{high}} \rightarrow \text{lower}$
 • Monotonically decreasing

Two state system:
 Ferromagnet:

$\uparrow \downarrow \uparrow \dots$
 $\underbrace{\hspace{10em}}_{N \text{ total}}$

$N_{\uparrow}, N_{\downarrow}$

$$N = N_{\uparrow} + N_{\downarrow}$$

$$\bullet \binom{N}{N_{\uparrow}} = C(N, N_{\uparrow}) = \frac{N!}{N_{\uparrow}! (N - N_{\uparrow})!} = \frac{N!}{N_{\uparrow}! N_{\downarrow}!}$$

$$\bullet p(N_{\uparrow}) = \frac{W(N_{\uparrow})}{W_{\text{tot}}}$$

$W_{\text{tot}} = \text{adds all possible outcomes}$

$$W(1) = \binom{N}{1}, \quad W(3) = \binom{N}{3}$$

$$W_{\text{tot}} = \sum_{n=1}^N \binom{N}{n}$$

$$\ast W_{\text{tot}} : 2^N$$

$$p(N_{\uparrow}) = \frac{N!}{2^N N_{\uparrow}! N_{\downarrow}!}$$

$$\sum_{n=1}^{\infty} -\gamma n^2$$

$$\sum_{n=1}^{\infty} E_n$$

Grand Canonical Ensemble:

$$S_t(E_{tot}) \geq S_1(E_1) + S_2(E_2)$$

$$S_t(E_{tot}) \equiv k_B \ln C$$

$W_1 \cdot W_2$: Before interactions

Micro $\longrightarrow$ Macro
QM Thermo...

Micro states: Counts accessible
micro states

$$\hat{H} |\psi_{\text{system}}\rangle = E |\psi_{\text{system}}\rangle$$

$|\psi_{\text{system}}\rangle$: Too complicated / Not relevant.

QM: Guides our thinking

• Observables in QM

$$\begin{aligned} \text{Ex.: } \langle \hat{H} \rangle &= \langle E \rangle = \sum_n \langle n | \hat{H} | n \rangle \\ \langle \hat{p} \rangle &= \sum_n \langle n | \hat{p} | n \rangle \quad \text{moments} \\ \text{etc.} \dots \\ \hat{H} &= \sum_i \frac{\hat{p}_i^2}{2m_i} + V(q) \end{aligned}$$

So, usually assume $V(q) \approx 0$ initially:
So, looking at

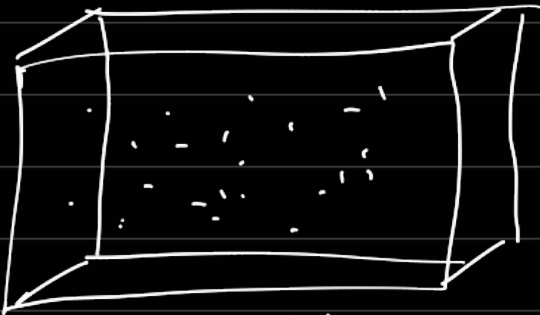
$$H \longleftrightarrow P$$

Momenta & Energy main interests

• Fundamental Assumption...

example:

Gas in a box



$$\bullet \langle \hat{H} \rangle = \sum_i \langle \hat{p}^2 / 2m \rangle$$

~> Momenta $\rightarrow$ Velocity

• Discrete energy levels for velocity

$$E_i = \frac{\hbar^2 \pi^2}{2mL^2} (n_x^2 + n_y^2 + n_z^2) = \frac{p^2}{2m}$$

$p = \dots$

$W(E_i) \equiv \#$ of microstates with energy E

$$\bullet n_x = n_y = 0, n_z = 5$$

$$E \sim 25 \gamma$$

$$\bullet n_z = 0, n_x = 3, n_y = 4$$

$$9 + 16 = 25$$

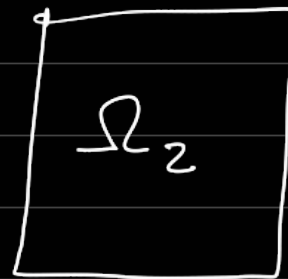
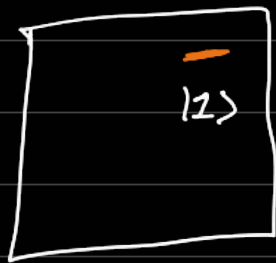
$$P(E) = \frac{1}{W(E)}$$

* Micro-concave:

Here we have an isolated system "ideal"
case $E = \text{const.}$
 $V_0 = \text{const.}$

Counting problem:

- En. eigen state



$|4\rangle_2$

$$P(1, 2) = P(2) P(1) = \frac{1}{\Omega_1 \Omega_2} =$$

$$\Omega_{\text{tot}} = \Omega_{12} = \Omega_1 \Omega_2$$

Also system 1:

1, 2	2 = Ω_1
z:	1, 2, 3
	3 = Ω_2

$$\left\{ \begin{array}{l} (1, 1), (1, 2), (1, 3) \\ (2, 1), (2, 2), (2, 3) \end{array} \right\} \frac{6}{3}$$

$$W(E_1, E_2) = W(E_1) W(E_2) \quad (\text{non interacting})$$

- Maximizing W_{12} maximizes P :

$$\frac{1}{T} = \frac{\partial S}{\partial E}$$

* Idea: $\Omega(E_1, E_2) \xrightarrow{\text{over time}} \tilde{\Omega}_{\text{tot}}$

- Energies are further split: so more eigenstates

energy, states available for each state
 → Increasing V also immediately increases Ω

$$\tilde{\Omega}_{\text{tot}} \geq \Omega_{12}$$

$$S(E_{\text{tot}}) \equiv k_B \ln(\tilde{\Omega}_{\text{tot}}) \geq$$

↳ Assumed at equilibrium

- Now, the claim is that $(\Omega_{12})^* \approx \tilde{\Omega}_{\text{tot}}$

$$\frac{\partial}{\partial E}(W_1 W_2) = \frac{\partial S_1}{\partial E} \Big|_{E_*} = \frac{\partial S_2}{\partial E} \Big|_{E_*}$$

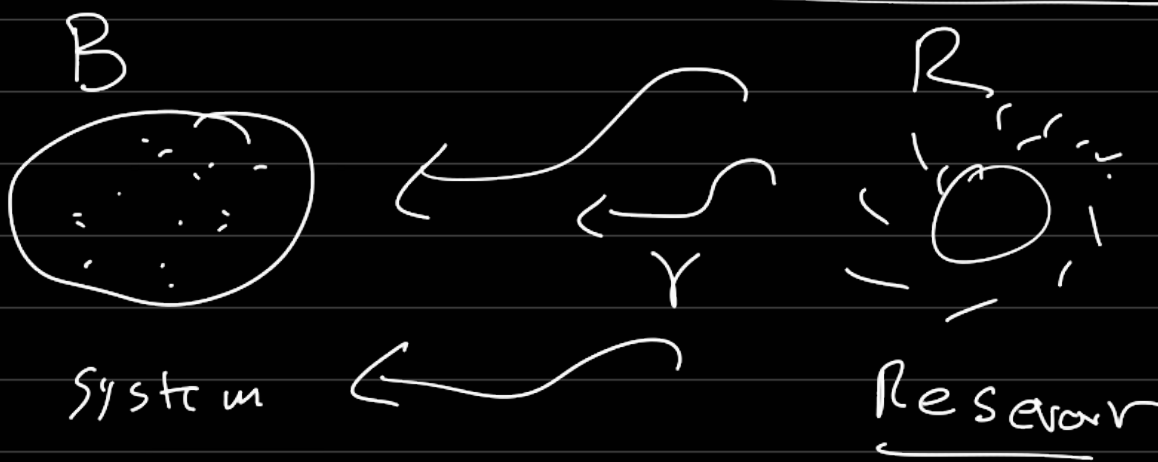
$$E_* : S_{E_*} \geq 2S_{\text{all}}$$

$$\frac{\Omega_E}{\Omega_{\text{all}}} = \frac{e^{S_*}}{e^{S_{\text{all}}}} = \frac{e^{2N}}{e^N}$$

$$\frac{\Omega_{E_*}}{\Omega_{\text{all}}} = \frac{e^N}{e^N} \leftarrow 10^{23}$$

Micro : (E, N) : fixed

• Idea: fix temp: deduce average en.



(Heat Bath)

Equilibrium temp T

Reservoir:

$$E_R \gg E_S$$

B: $|n\rangle$ s.t. $\hat{H}|n\rangle = E_n|n\rangle$

$$P_n = \frac{\Omega_R(E_R = E_{\text{tot}} - E_n)}{\Omega_{\text{tot}}}$$

Ω_n : # of microstates of R associated with a single microstate n of Box

$$\left(\frac{S(E_{\text{tot}} - E_n)}{k_B} \right)$$

$$\Omega_R(E_{\text{tot}} - E_n) = e$$

$$E_{\text{tot}} \gg E_n$$

$$S(E_{\text{tot}} - E_n) = S_{\text{tot}} - \underbrace{\frac{\partial S}{\partial E}}_{\frac{1}{T_0}} E_n + \dots$$

$$\Omega_R(E_{\text{tot}} - E_n) = e^{\frac{S_{\text{tot}}}{k_B}} \cdot e^{-E_n/k_B T}$$

$$\Omega_{\text{tot}} = \sum_n e^{\frac{S_{\text{tot}}}{k_B}} e^{-E_n/k_B T}$$

$$P_n = \frac{e^{-E_n/k_B T}}{\sum_n e^{-E_n/k_B T}}$$

$$Z \equiv \sum_n e^{-E_n/k_B T} \quad (\text{Partition function})$$

